

## Customer Journey Map

|               |                                                              |
|---------------|--------------------------------------------------------------|
| Date          | 14 July 2026                                                 |
| Team ID       |                                                              |
| Project Name  | OptiCrop – Smart Agricultural Production Optimization Engine |
| Maximum Marks | 2 Marks                                                      |

### Customer Journey Map

Mapping out the customer's experience stage-by-stage, capturing their actions, touchpoints, thoughts, and feelings, along with process ownership and improvement opportunities at each stage.

| Phase of Journey  | Stage1                                                                                                             | Stage2                                                                                                          | Stage3                                                                                                               |
|-------------------|--------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------|
| Actions           | User opens the OptiCrop web application and enters soil parameters (N, P, K, Temperature, Humidity, pH, Rainfall). | The Flask application sends the input to the trained Random Forest model, which predicts the most suitable crop | The system displays the recommended crop along with crop image, growing season, water requirement, and farming tips. |
| Touch point       | OptiCrop Home Page / Find Crop Page                                                                                | Flask Backend & Machine Learning Model                                                                          | Result Page                                                                                                          |
| Customer Thought  | "I hope the system recommends the best crop for my field."                                                         | "The prediction is being generated based on my soil data."                                                      | "Now I know which crop is suitable for my land."                                                                     |
| Customer Feeling  | Curious and hopeful                                                                                                | Interested and confident                                                                                        | Satisfied and informed                                                                                               |
| Process Ownership | User Interface (Frontend)                                                                                          | Flask Backend & Machine Learning Model                                                                          | Result Display Module                                                                                                |

|                           |                                                       |                                                                |                                                                                   |
|---------------------------|-------------------------------------------------------|----------------------------------------------------------------|-----------------------------------------------------------------------------------|
| Improvement Opportunities | Provide input validation and tooltips for each field. | Improve prediction speed and integrate real-time weather data. | Add fertilizer recommendations, nearby market prices, and crop disease detection. |
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